

Name: _____

Lab Exercise: Preparing a Smear

Bacteria have almost the same refractive index as water. This means that when you try to view them using a microscope they appear as faint, gray shapes and are difficult to see. Staining cells makes them easier to visualize.

Three Steps to Stain: Smear, Fix, Stain

1. Prepare a Smear - A properly prepared smear accomplishes two things. It causes bacteria to adhere to a slide so that they can be stained and observed. It also kills them, rendering pathogenic bacteria safe to handle. When learning to prepare smears, it is important to recognize the correct density of bacteria to place on the slide. Too many, and they overlap each other giving false positives or crowding each other making shape identification difficult. Too few, and they cannot be located on the slide.

a. Smear - To prepare a smear from a:

i. dry culture –

1. Sterilize a loop.
2. Pass the top of tube of distilled water through the flame to decrease airborne contaminants.
3. Insert loop into tube to collect a loopful of distilled water. Transfer this loop onto the center of a clean microscope slide.
4. Sterilize the loop again.
5. Aseptically remove a small amount of material from a culture. Mix the bacteria on the loop with the drop of water on the slide. Smear this mixture along the surface of the slide to spread it very thinly.
6. Allow the surface of the slide to air dry.

ii. broth culture –

1. Place a drop of broth culture directly on the slide without adding water.
2. Smear the drop along the surface of the slide to spread it very thinly.
3. Allow the surface of the slide to air dry.

b. Fix –

- i. Quickly pass the slide through a flame. Three quick passes are usually sufficient. The slide should be cool enough to touch to the back of your hand without burning.

c. Stain –

- i. After cooling the slide, the specific staining procedure is conducted.